# Homework 5 – Theory

## 1

### (a)

From $\Sigma$,

$$\mathrm{Cov}(X_3, X_4) = \Sigma_{34} = 0.$$

Thus, $\boxed{X_3 \text{ and } X_4 \text{ are not correlated.}}$

### (b)

Using the $\{3, 4\}$ block of $Q$,

$$\mathrm{Cov}((X_3, X_4) \mid X_1, X_2) = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}^{-1} = \begin{bmatrix} 0.2 & 0 \\ 0 & 0.2 \end{bmatrix}.$$

$$\boxed{\mathrm{Cov}(X_3, X_4 \mid X_1, X_2) = 0}$$

### (c)

The only nonzero off-diagonal entry in row 2 of $Q$ is $Q_{21} = 3$.

$$\boxed{M_2 = \{X_1\}}$$

### (d)

Using $Y_1 = X_1 + X_4$ and $Y_2 = X_2 - X_4$,

$$\begin{aligned} \mathrm{Var}(Y_1) &= \Sigma_{11} + 2\Sigma_{14} + \Sigma_{44} \\ &= 0.71 + 0 + 0.2 = 0.91, \\ \mathrm{Var}(Y_2) &= \Sigma_{22} - 2\Sigma_{24} + \Sigma_{44} \\ &= 0.46 - 0 + 0.2 = 0.66, \\ \mathrm{Cov}(Y_1, Y_2) &= \Sigma_{12} - \Sigma_{14} + \Sigma_{42} - \Sigma_{44} \\ &= -0.43 - 0 + 0 - 0.2 = -0.63. \end{aligned}$$

$$\boxed{\Sigma_Y = \begin{bmatrix} 0.91 & -0.63 \\ -0.63 & 0.66 \end{bmatrix}}$$

# 2

## (a)

E-step:

$$p = \frac{e^{-1/2}}{e^{-1/2} + e^{-9/2}} = \frac{1}{1 + e^{-4}} \approx 0.98.$$

By symmetry, the other component has the same probability for $x = 1$.

M-step:

$$\mu_1 = \frac{p(-1) + (1-p)(1)}{p + (1-p)} = 1 - 2p \approx -0.96,$$
$$\mu_2 = \frac{(1-p)(-1) + p(1)}{(1-p) + p} = 2p - 1 \approx 0.96.$$

$$\boxed{\mu_1 \approx -0.96, \qquad \mu_2 \approx 0.96}$$

## (b)

No. By symmetry, let $(\mu_1, \mu_2) = (-a, a)$ with $a > 0$.

$$p = \frac{1}{1 + e^{-2a}}, \qquad a_{\text{new}} = 2p - 1 = \frac{e^{2a} - 1}{e^{2a} + 1} = \tanh(a).$$

Since $0 < \tanh(a) < a$, the sequence decreases to $L$ satisfying $L = \tanh(L)$. Thus, $L = 0$.

$$\boxed{(\mu_1, \mu_2) \longrightarrow (0, 0)}$$

## (c)

Since $\mu_1 = \mu_2 = 2$, both components have equal probabilities.

$$\mu_1 = \mu_2 = \frac{\frac{1}{2}(-1) + \frac{1}{2}(1)}{\frac{1}{2} + \frac{1}{2}} = 0.$$

The means remain at zero.

$$\boxed{(\mu_1, \mu_2) = (0, 0)}$$

## (d)

Each point is assigned to its nearest center:

$$-1 \to \mu_1, \qquad 1 \to \mu_2.$$

The updated centers are

$$\mu_1 = -1, \qquad \mu_2 = 1.$$

The assignments remain unchanged.

$$\boxed{(\mu_1, \mu_2) = (-1, 1)}$$